MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6310

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex oscillator, integral cavity, tunable frequency range 8500 to 10,000 MHz

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Irs	Tuner plate	tk	F	Alt
Unit:	V	Vdc	Vdc	mAdc	T°C	(See note 1)	MHz	ft
Maximum:	7.0	350	-20 to -1,000	42	200	--	10,000	10,000
Minimum:	5.7	---	---	--	---	--	8,500	---

PHYSICAL CHARACTERISTICS: (See outline drawing)

TEST CONDITIONS:

Parameter:	Ef	Ers	Er	Tuner plate temperatures	F
Unit:	V	Vdc	Vdc	°C	MHz
	6.3	300	-55 to -225	(See notes 2 and 3)	9,300 ±0.3%

REQUIREMENT OR TEST:

GENERAL

Qualification - Required

6310

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification					
1031	Operation vibration (1)	Er(Mode 5)/max Po; 10G, F = 50 to 600 Hz; t = 5 min	ΔF(p-p)	---	4	MHz
----	Shock	Er(Mode 5)/max Po; G = 200; shock duration = 0.001 sec See notes 5 and 8	ΔF	---	5	MHz
4280	Electronic tuning range (2)	Er(Mode 5)/50% max Po; F = 8,500 to 10,000 MHz	ΔF(p-p)	30	---	MHz
4027	Temperature coefficient	Er(Mode 5)/max Po; TA = 20° to 60°C	ΔF/Δ°C	0.0	-0.20	MHz/°C
----	Frequency modulation	Er(Mode 5)/max Po; Ef = 5.7 to 7.0 Vdc	ΔF	---	0.1	MHz
----	Heater voltage coefficient	Ef = 5.7 to 7.0 V	ΔF/ΔEf	---	6	MHz/v
4223	Mechanical tuning rate	F = 8,500 to 10,000 MHz	ΔF/ΔTuner degrees	1.8	3	MHz/degree
4250	Power output (3)	Er(Mode 5)/max Po; F = 8,500 to 10,000 MHz	Po	25	---	mW
	Quality conformance inspection, part 1					
1031	Operation vibration (2)	10G, F = 50, t = 120 See note 4	Ir	0	10	μAdc
4229	Reflector current	See notes 6 and 7	Ir	---	3	μAdc
4214	Emission	Ef = 5.7 See note 7	ΔIrs/Irs	---	-15	%
4212	Resonator current	Er(Mode 5)/max Po;	Irs	20	32	mAdc
4250	Power output (1)	Er(Mode 5)/max Po;	Po	25	---	mW
4223	Mechanical tuning range	Er(Mode 5)/max Po; high frequency low frequency	F F	10,000 ---	--- 8,500	MHz MHz
4213	Reflector voltage (1)	Er(Mode 5)/max Po	Er	-115	-180	Vdc
----	Broadband testing	Er(Mode 5)/max Po; at = 8 db min; See note 9 VSWR = 2.5/1 max.				

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2					
1301	Heater current		If	1.08	1.32	A
4250	Power output (2)	Er(Mode 5)/max Po; F = 8,500 ±0.3% MHz; F = 10,000 ±0.3% MHz	Po Po	25 25	-- --	mW mW
4213	Reflector voltage (2)	Er(Mode 5)/max Po; F = 8,500 ±0.3% MHz; F = 10,000 ±0.3% MHz	Er Er	-85 -160	-135 -225	Vdc Vdc
4280	Electronic tuning range (1)	Er(Mode 5)/50% max Po	ΔF	30	--	MHz
4231	Electronic tuning hysteresis	Er(Mode 5)/max Po	Ratio	--	0.5	---
	Quality conformance inspection, part 3					
4551	Life-test	Group C, power output (1) conditions	t	500	--	hr
4551	Life-test end points:					
4250	Power output (1)	Er(Mode 5)/max Po	Po	20	--	mW
4229	Reflector current	T = 5 min See notes 6 and 7	Ir	--	10	μAdc

NOTES:

1. No warmup time is required in order to obtain the specified life from the tube. If extended life is desired, a heater warmup period of approximately 1 minute is recommended before the application of the resonator voltage.
2. All oscillation tests except vibration tests shall be made with the tube rigidly connected to a UG39/U flange on appropriate RG52/U waveguide equipment and the load VSWR for the tube shall be less then 1.1. Forced-air cooling is required for power inputs above 10 watts.
3. The temperature of base and cap of the tube shall not exceed 120°C.
4. The reflector current shall be recorded with a Brush Model BL202 recorder, or equivalent. There shall be no reflector current bursts greater than the limit shown.
5. The tube shall be given five shocks in each of three planes. The frequency shift after shock in one plane shall not exceed the value specified.
6. After 2 minutes with all voltages applied, total reflector current shall not exceed the specified limits.
7. The tube shall not be oscillating during the test.
8. Connection to the tube shall be made with an Amphenol 78-S3S-1001-3 pin connector and a miniature type reflector cap connector, or equivalents. Flexible leads shall be used in making connections to the tube.

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NOTES: -Continued

9. Superimposing sufficient 60-hertz a c voltage on the direct reflector voltage to suppress oscillations on the ends of the sweep, the mode curve shall be observed on an oscilloscope whose amplifier has minimum bandpass of 0.1 MHz. With a standing wave introducer inserted, there shall be no discontinuities at the maximum power point for any phase of the standing wave. If the center of the mode is distorted, a wavemeter pip shall not disappear for an interval of the mode center as the wavemeter is tuned to check for a frequency discontinuity.

Custodians:

Army - EL
Navy - EC
Air Force - 85

Review activities:

Army - EL
Navy - SH
Air Force - 80, 84
DSA - ES

User activities:

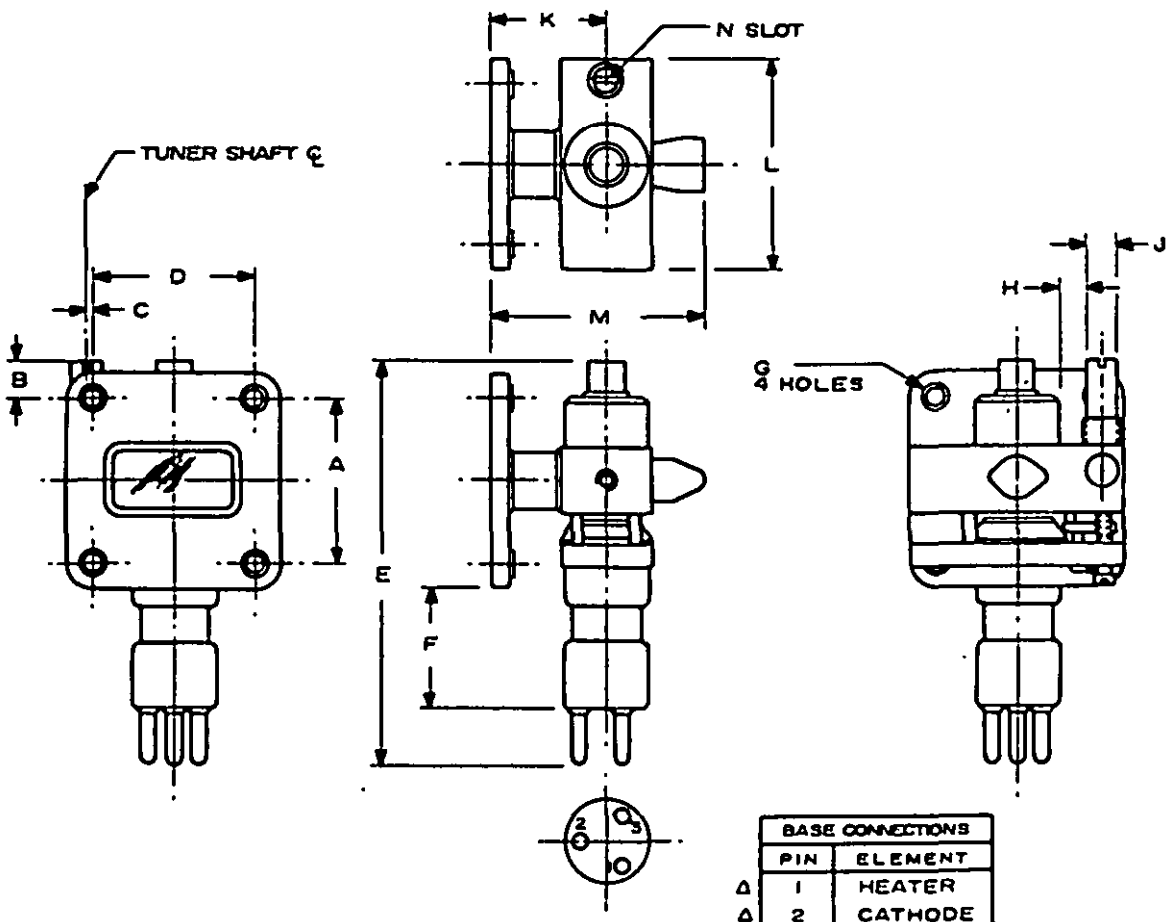
Army - MU
Navy - MC, CG
Air Force - 11

Preparing activity:

Air Force - 85

(Project No. 5960-2853)

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BASE CONNECTIONS	
PIN	ELEMENT
Δ 1	HEATER
Δ 2	CATHODE || 3	HEATER

Δ 1 & 2 INTERNALLY CONNECTED

DIMENSIONS								
LTR	INCHES		MILLIMETERS		LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
QUALIFICATION					QUALITY CONFORMANCE INSPECTION, PART 2				
F		1.000		25.40	J	.248	.250	6.30	6.35
H	.160		4.06		K	.850	.936	21.59	23.77
L		1.750		44.45	E		3.125		79.38
QUALITY CONFORMANCE INSPECTION, PART 1					NOMINAL DIMENSIONS				
A	1.276	1.284	32.41	32.61	B	.187		4.75	
D	1.216	1.224	30.89	31.09	G	.219 .185 (REMOVABLE INSERT)		5.56 4.70	
M		1.750		44.45					
QUALITY CONFORMANCE INSPECTION, PART 2					N	.050 WIDE .220 DEEP		1.27 5.59	
C	.010	.070	.25	1.78					

NOTES:
1. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm.
2. NOMINAL DIMENSIONS ARE FOR INFORMATION ONLY AND ARE NOT REQUIRED FOR INSPECTION PURPOSES.
3. BASE: PEEWEE 3 PIN, A3-1
4. CAP: C1-4

FIGURE 1. Outline drawing of type 6310

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